

PHISHING DETECTION & AWARENESS REPORT

Future Interns – Cyber Security Task 2 (2026)

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Project: Phishing Email Detection & Awareness System

1. Introduction

Phishing is a social-engineering attack in which attackers attempt to deceive users into clicking unsafe links, opening malicious attachments, or providing sensitive information such as passwords and verification codes. The purpose of this project is to analyze representative phishing samples and translate the findings into practical awareness guidance for users.

This report documents three investigation cases. The first two cases focus on phishing email examples, while the third is an SMS phishing (smishing) example included as an additional awareness case. Each case is reviewed for suspicious indicators, social-engineering techniques, and overall risk.

2. Project Objectives

- Analyze publicly available phishing message samples.
- Identify visible phishing indicators and suspicious characteristics.
- Classify the overall risk of each sample.
- Explain the attack technique in clear, non-technical language.
- Provide practical prevention and awareness guidance.

3. Tools and Resources Used

- Public phishing examples repository for educational sample review.
- GitHub for accessing sample screenshots and accompanying analysis.
- VirusTotal findings documented by the public sample repository for the malicious attachment case.
- Microsoft Word for documentation and report preparation.

Suspicious links and attachments were treated as evidence only and were not intentionally clicked or executed during this project.

4. Analysis Methodology

The following workflow was used for each case:

1. Collect and preserve the public sample.
2. Identify available sender, subject, message body, links, or attachments.
3. Review accompanying public analysis and visible indicators.
4. Identify social-engineering techniques.
5. Assess the potential user impact.
6. Assign a final classification and risk level.
7. Document recommended safe actions.

5. Case 001 – Microsoft Password Expiration Phishing

5.1 Case Overview

Case ID: FUTURE-CS02-001

Attack Type: Credential Phishing / Brand Impersonation

Risk Classification: PHISHING – HIGH RISK

The sample impersonates a trusted Microsoft-related identity and uses a password-expiration message to encourage the recipient to interact with an embedded action button.

Claimed Sender Display Name: IT STAFF

Subject: Not visible in the provided sample screenshot

Main Theme: Password expiration

Suspicious Action: "Keep Same Password"

5.2 Identified Indicators

- Possible imitation or misuse of Microsoft branding.
- Unusual recipient-name formatting in the greeting "Dear sean,".
- Writing and capitalization inconsistency in "Your Password".
- Logically suspicious call-to-action: an expired password message paired with a "Keep Same Password" button.
- Password-related social engineering intended to encourage user interaction.

5.3 Attack Technique

The attack uses brand impersonation and a security-related account message. By presenting a password problem, the message attempts to make the recipient believe that action is required. The suspicious action button is the primary interaction point and could expose the recipient to further credential-phishing activity.

5.4 Risk Assessment

Indicator	Risk Level
Brand imitation	High
Unusual name formatting	Medium
Writing inconsistency	Low
Suspicious password-related action button	High

Final Classification: PHISHING – HIGH RISK

Recommended Action: Do not click the action button. Do not enter credentials. Verify password notifications independently through official channels and report suspicious messages to the relevant IT or security team.

6. Case 002 – Apple App Store Invoice Phishing

6.1 Case Overview

Case ID: FUTURE-CS02-002

Attack Type: Malicious Attachment / Brand Impersonation

Risk Classification: PHISHING – HIGH RISK

This sample portrays itself as an Apple App Store-related invoice or purchase confirmation and attempts to persuade the recipient to open an attached PDF.

Displayed Sender: App Store

Displayed Sender Address: billingservice-info003@refundsbweb.com

Main Theme: Mobile application purchase or invoice confirmation

Suspicious Attachment: Invoice.pdf

6.2 Identified Indicators

- The displayed sender address does not appear to be an official Apple domain.
- An unexpected PDF attachment is included and the recipient is encouraged to open it.
- The purchase-notification theme may create curiosity or concern about an unrecognized transaction.
- The wording "Please open the attached (PDF) file" contains unusual formatting and punctuation.
- The public sample documentation includes a VirusTotal result indicating that the attachment was flagged as malicious.

6.3 Attack Technique

The attack combines brand impersonation, a fake purchase or invoice scenario, and a malicious attachment. The recipient may be motivated to open the attachment to investigate an unfamiliar transaction.

6.4 Risk Assessment

Indicator	Risk Level
Suspicious sender address	High
Unexpected PDF attachment	High
Fake purchase notification	Medium
Writing inconsistency	Low
Documented malicious attachment finding	High

Final Classification: PHISHING – HIGH RISK

Recommended Action: Do not open unexpected attachments. Verify purchase notifications through the official account or application. Report suspicious emails and follow organizational security procedures for potentially malicious files.

7. Case 003 – Coinbase SMS Phishing (Smishing Awareness Case)

7.1 Case Overview

Case ID: FUTURE-CS02-003

Attack Type: Smishing (SMS Phishing) / Brand Impersonation

Risk Classification: PHISHING – HIGH RISK

This case is an additional awareness example involving a public SMS phishing sample that impersonates the Coinbase brand and directs the recipient toward a deceptive Coinbase-related link.

7.2 Identified Indicators

- Use of a trusted brand name to increase credibility.
- A suspicious link that appears related to Coinbase.
- A visually similar but different character used within the deceptive domain name.
- A homograph-style character-substitution technique intended to make the malicious domain appear legitimate.
- Security-related social engineering that may pressure the recipient to act without carefully checking the link.

7.3 Attack Technique

This attack uses SMS phishing, also known as smishing. The attacker combines brand impersonation with a visually deceptive domain. The character substitution technique is particularly dangerous because the URL may appear legitimate during a quick visual inspection.

7.4 Risk Assessment

Indicator	Risk Level
Coinbase brand impersonation	High
Suspicious link	High
Deceptive character substitution	High
Security-related social engineering	Medium

Final Classification: PHISHING – HIGH RISK

Recommended Action: Do not click unexpected SMS links. Carefully inspect domains, access financial services through official applications or manually typed addresses, and never provide passwords or verification codes after following an unexpected link.

8. Cross-Case Phishing Indicators Summary

Indicator	Case 001	Case 002	Case 003
Brand impersonation	Yes	Yes	Yes
Suspicious link/action	Yes	Attachment-focused	Yes
Suspicious attachment	No	Yes	No
Social engineering	Yes	Yes	Yes
Writing/formatting anomaly	Yes	Yes	Not primary indicator

Deceptive domain/character	Not visible	Sender domain concern	Yes
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9. Risk Classification Framework

A simple educational risk model was used to support consistent classification. This is a project-specific awareness framework and not an official industry scoring standard.

Classification	Meaning
Safe	No significant suspicious indicators identified in the reviewed sample.
Suspicious	Some warning signs exist and independent verification is recommended.
Phishing	Multiple indicators or strong evidence suggest malicious intent.

All three cases in this report were classified as Phishing – High Risk because each contained multiple indicators associated with deception, impersonation, malicious interaction, or deceptive URLs/attachments.

10. Prevention and Awareness Guidelines

- Do not trust a brand logo or display name by itself.
- Check the full sender address and domain where available.
- Be cautious of unexpected password, account, payment, or security alerts.
- Do not click suspicious links or buttons.
- Inspect URLs carefully for misspellings and visually similar characters.
- Do not open unexpected attachments, especially when they create urgency or concern.
- Verify requests by visiting the official website or application independently.
- Do not share passwords, OTPs, or verification codes through links received unexpectedly.
- Report suspicious messages to the appropriate IT or security team.

11. Employee Do's and Don'ts

Do

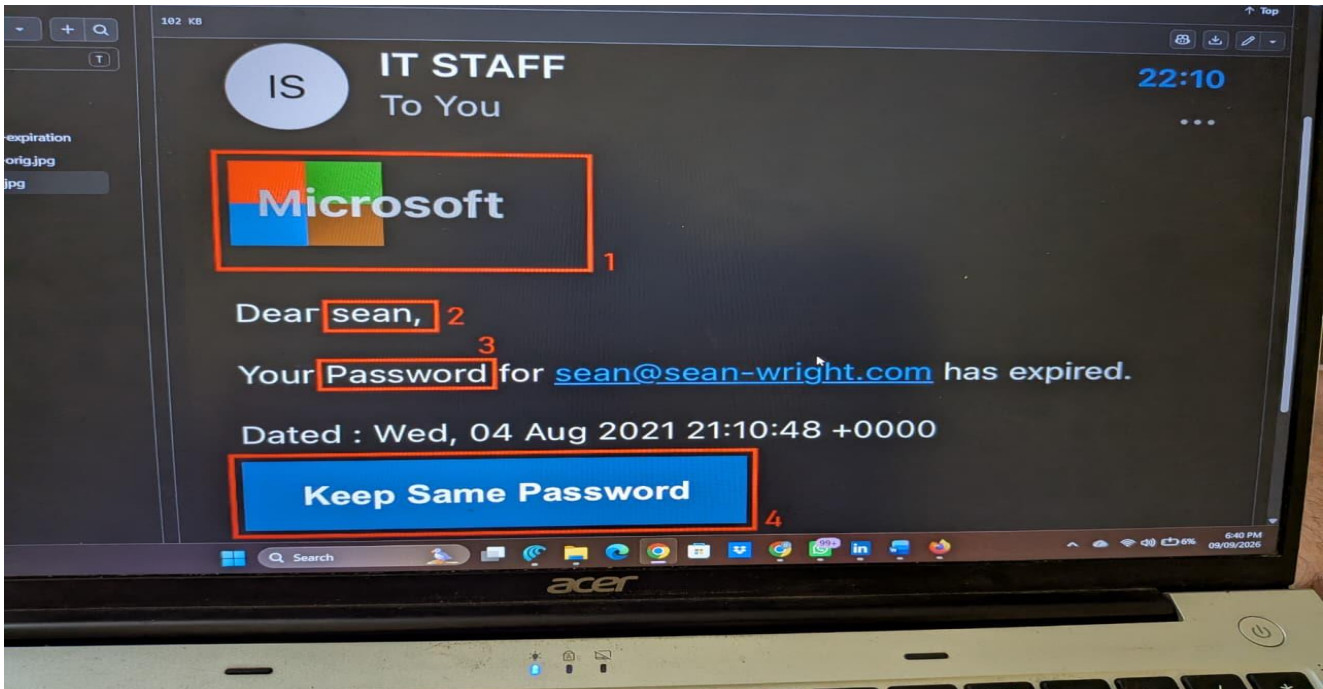
- Verify unexpected messages independently.
- Use official websites and applications for account actions.
- Report suspicious emails and messages promptly.
- Check sender addresses, domains, URLs, and attachments carefully.

Don't

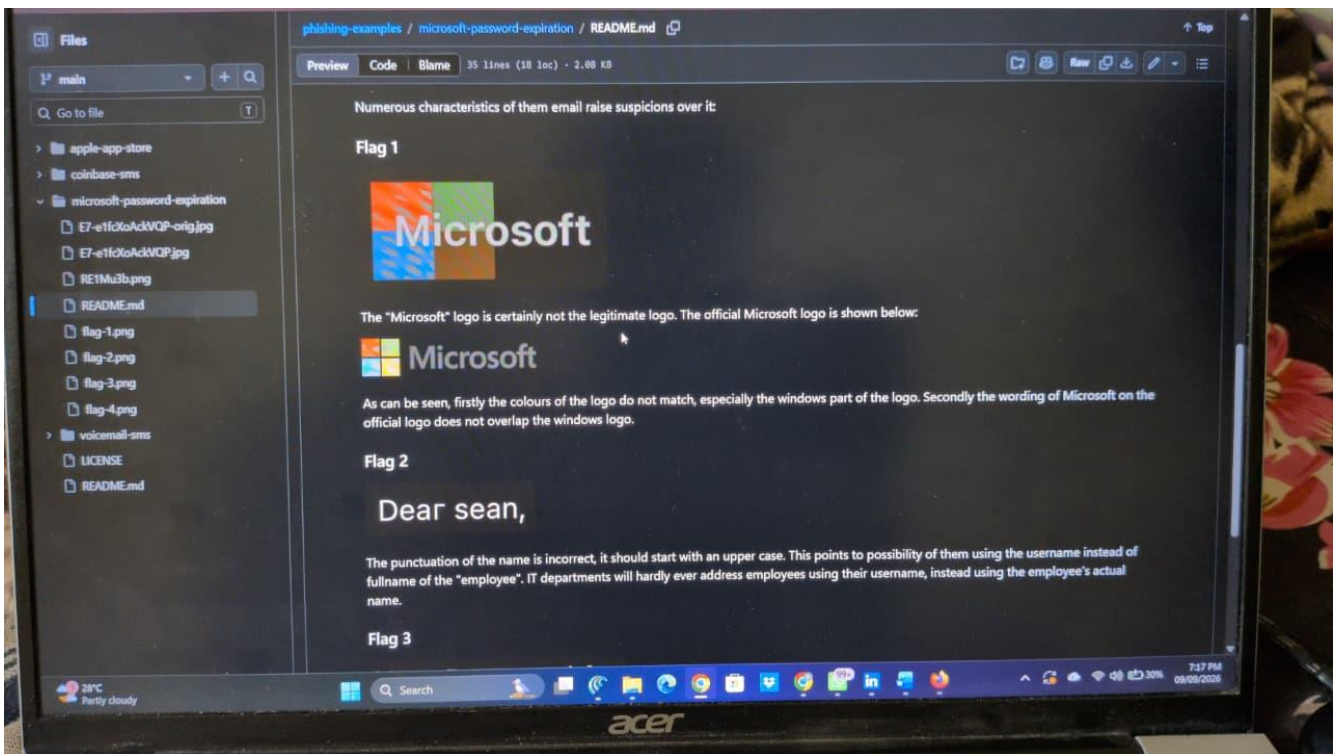
- Do not click links simply because a message appears urgent.
- Do not open unexpected attachments.
- Do not trust a message solely because it uses a familiar brand.
- Do not provide passwords, OTPs, or sensitive information through unexpected requests.

12. Evidence Appendix

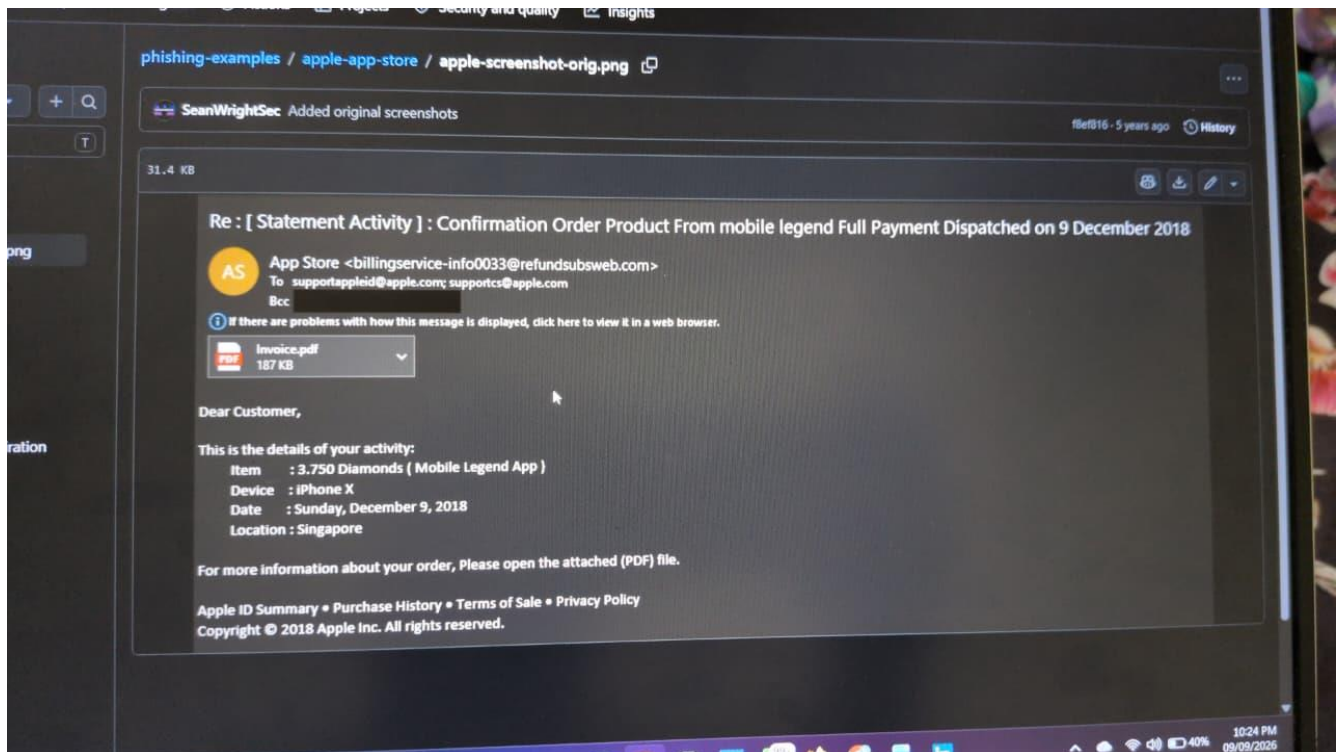
Case 001 – Original phishing email sample:



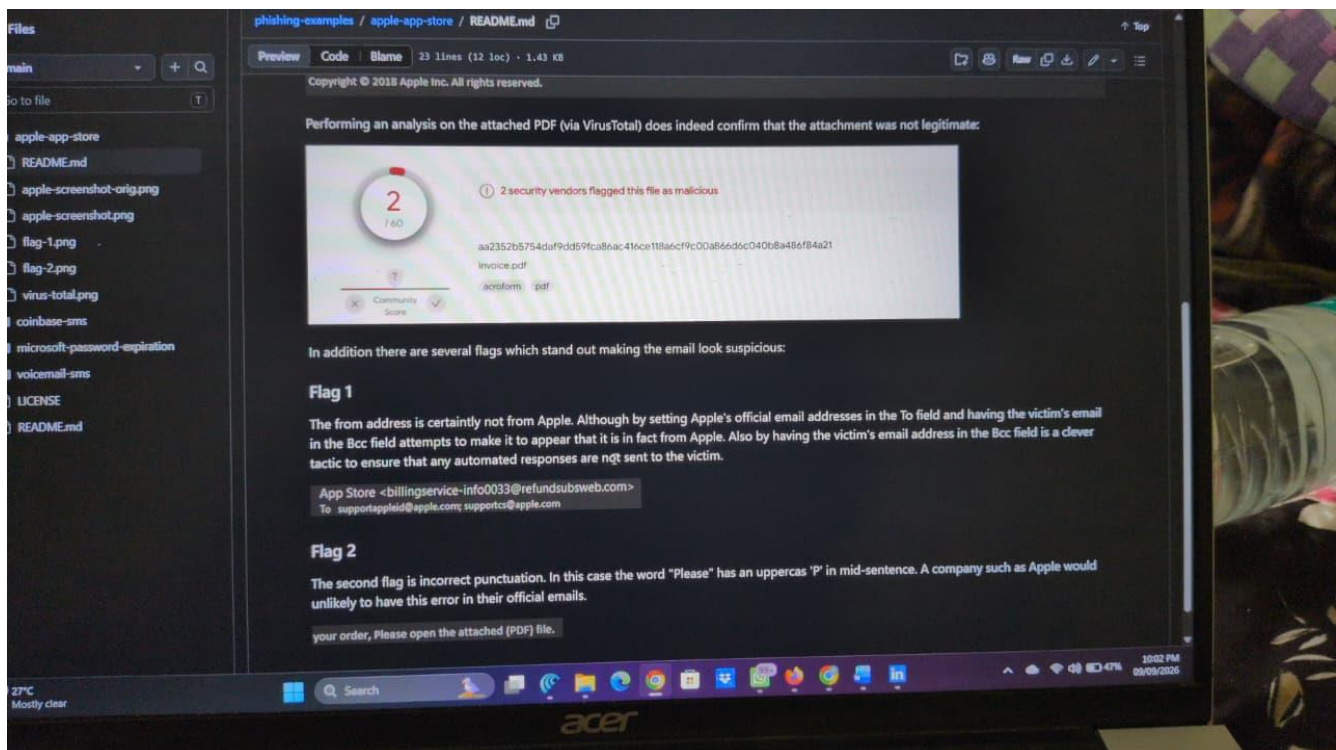
Case 001 – Flag analysis evidence:



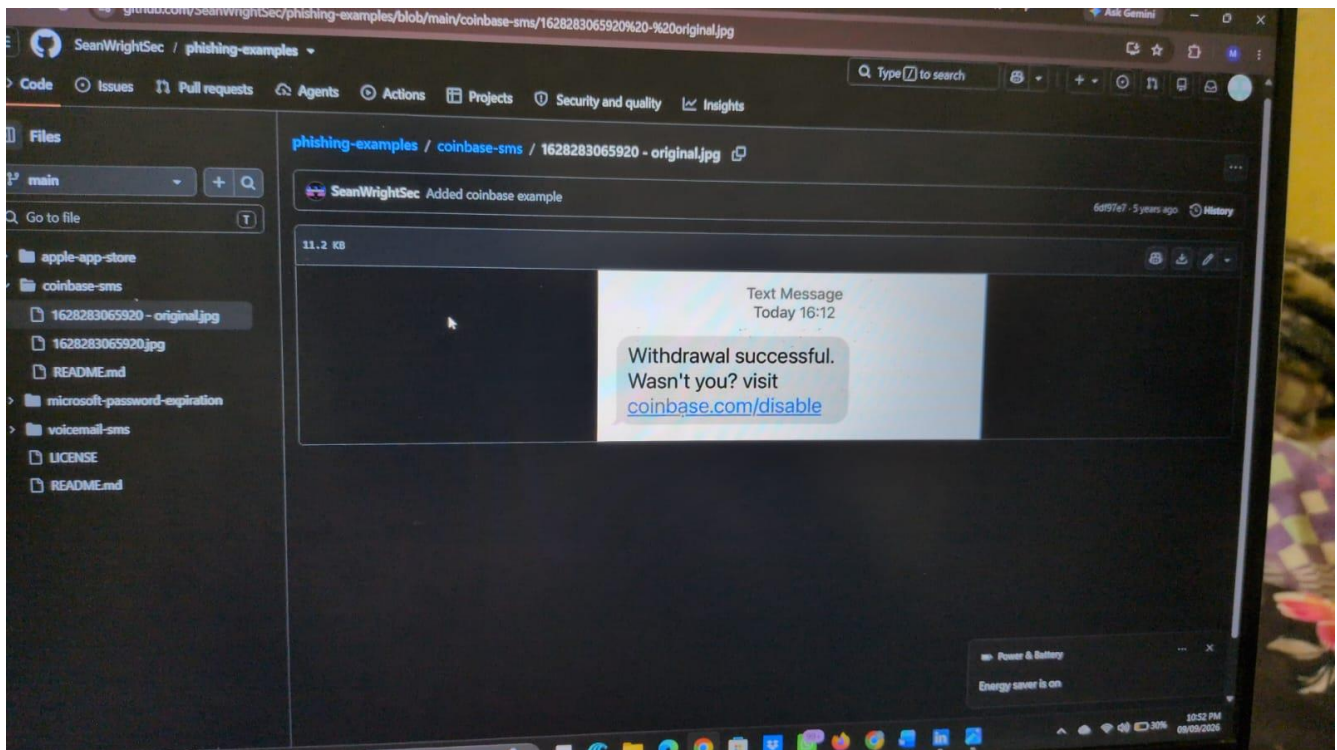
Case 002 – Original phishing email sample:



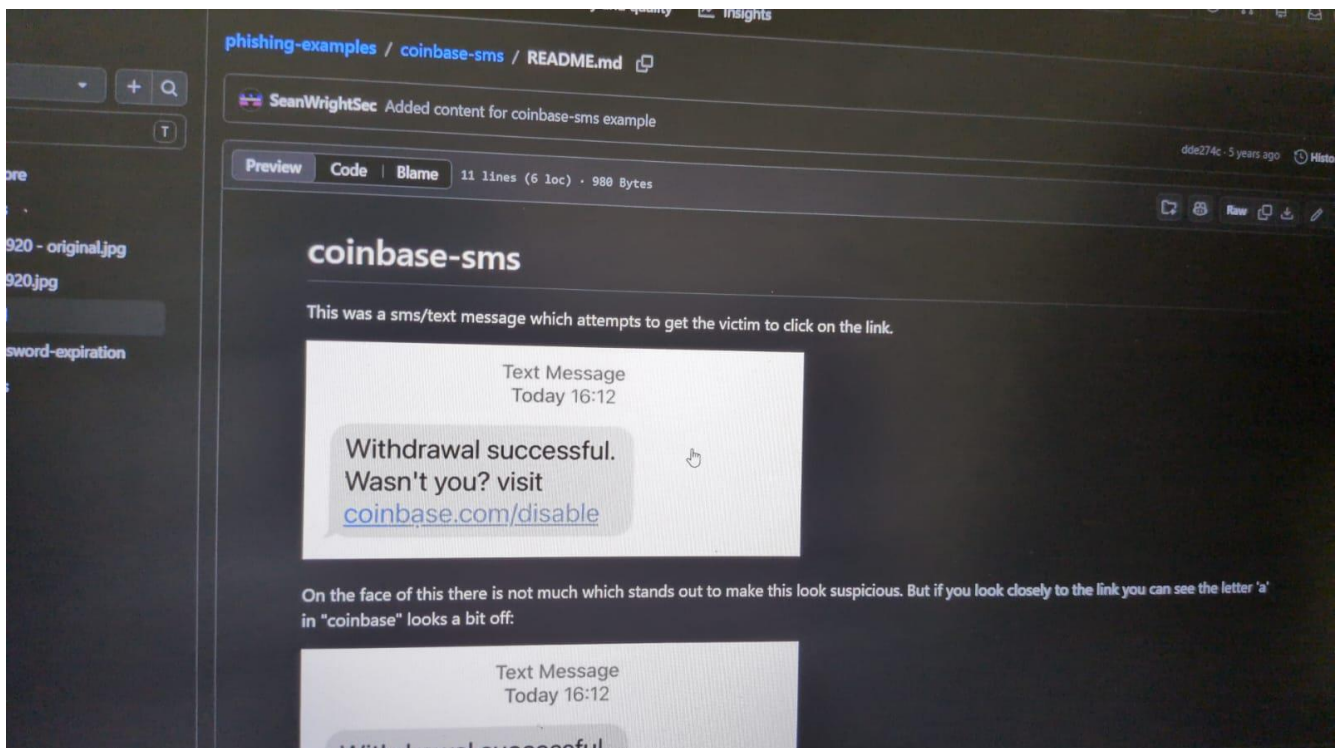
Case 002 – Phishing indicator and attachment-analysis evidence:



Case 003 – Original Coinbase SMS sample:



Case 003 – Suspicious link and deceptive-character analysis evidence:



13. Conclusion

This project demonstrates how phishing attempts can use different delivery methods and social-engineering techniques while pursuing the same objective: persuading users to perform an unsafe action. Across the three cases, the main warning signs included brand impersonation, suspicious sender or domain information,

deceptive links, unexpected attachments, formatting inconsistencies, and security or transaction-related pressure.

The most important awareness lesson is that users should pause before acting. Independent verification through official channels is safer than responding directly to an unexpected message. Combining technical checks with user awareness can reduce the likelihood of successful phishing attacks.